

TANEGASHIMA, Japan

WMO: 478370

Lat: 30.7204N

Lon: 130.9830E

Elev: 123

StdP: 14.63

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	42.0	44.0	27.4	21.5	46.9	29.8	23.9	47.6	39.6	49.9	36.4	50.6	15.8	330	0.463

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.6	89.3	80.1	88.1	79.6	87.1	79.2	81.7	87.0	80.8	86.0	80.1	85.1	9.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.1	156.8	85.0	79.3	152.9	84.4	78.7	149.6	83.9	45.5	87.3	44.6	86.2	43.9	85.3	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 34.0	29.7	26.2	38.0	91.3	2.3	1.2	36.3	92.2	34.9	92.9	33.6	93.5	31.9	94.4		
(5)			33.2	82.4	1.9	1.1	31.9	83.2	30.8	83.8	29.7	84.4	28.3	85.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.9	52.9	54.2	58.1	64.3	70.4	75.3	82.1	83.2	79.6	72.8	64.8	56.7
	DBStd	11.27	5.03	5.49	4.97	4.16	3.13	3.72	2.56	1.96	3.16	4.03	5.00	5.45
	HDD50	54	26	17	3	0	0	0	0	0	0	0	0	8
	HDD65	1289	376	305	218	61	2	0	0	0	0	2	64	261
	CDD50	6604	115	134	255	430	634	760	994	1028	887	707	443	217
	CDD65	2363	0	2	5	40	171	310	529	563	437	244	57	5
	CDH74	19475	0	0	1	35	370	1732	5778	6548	3960	981	68	2
	CDH80	5442	0	0	0	0	10	252	1846	2342	917	73	2	0
Wind	WSAvg	11.4	14.8	13.7	13.3	11.7	10.0	9.9	9.8	9.6	9.4	10.1	10.8	13.9
Precipitation	PrecAvg	124.0	6.3	6.9	9.0	10.7	11.8	25.2	10.7	8.0	13.1	8.6	7.8	6.1
	PrecMax	160.5	15.2	17.5	16.9	19.5	20.3	41.7	28.3	20.0	40.7	26.5	14.5	13.5
	PrecMin	60.3	0.1	2.6	3.3	4.4	4.3	10.4	0.0	0.7	2.2	1.4	2.5	1.2
	PrecStd	19.9	2.8	3.3	3.3	4.4	5.0	7.8	7.7	5.0	8.7	6.1	3.7	2.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.0	71.0	72.5	77.1	81.3	87.0	90.4	91.5	89.2	84.3	79.1	72.0
		MCWB	62.9	64.4	64.6	69.4	72.9	79.4	80.4	81.3	80.0	76.4	72.2	65.1
	2%	DB	65.9	67.9	70.0	74.8	79.1	85.0	89.0	89.8	87.3	82.0	75.8	69.1
		MCWB	60.5	63.1	62.8	67.3	70.8	78.4	80.1	80.2	78.3	74.6	69.1	63.2
	5%	DB	63.3	65.5	68.0	72.9	77.4	82.9	87.9	88.6	86.0	80.2	73.9	67.2
		MCWB	57.8	60.6	62.1	66.0	69.7	77.3	79.7	79.7	77.5	73.5	67.3	61.3
	10%	DB	60.9	63.0	66.0	71.2	76.1	81.0	86.8	87.5	84.5	78.7	72.2	65.2
		MCWB	55.0	57.8	60.0	64.5	68.9	76.1	79.3	79.4	76.9	72.0	65.7	59.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.6	66.5	67.6	72.4	75.6	80.1	82.4	83.0	81.5	78.5	73.7	67.6
		MCDB	67.6	69.5	70.5	75.1	78.9	85.1	87.8	88.4	86.7	82.1	77.7	69.9
	2%	WB	61.5	63.8	65.4	70.1	74.1	78.9	81.4	81.9	80.1	76.8	71.0	64.6
		MCDB	64.8	67.1	68.4	72.9	77.2	83.3	86.6	87.4	85.0	80.6	74.6	68.2
	5%	WB	58.7	61.3	63.3	68.3	72.6	77.9	80.7	81.0	79.2	75.3	68.9	62.3
		MCDB	62.8	64.9	66.7	71.3	75.7	81.9	85.8	86.3	83.8	79.1	72.6	66.4
	10%	WB	55.6	58.6	61.1	66.1	71.0	77.0	80.0	80.3	78.3	73.6	67.1	59.7
		MCDB	60.2	62.5	64.9	69.8	74.2	80.6	84.9	85.5	82.8	77.5	71.1	64.4
Mean Daily Temperature Range	5% DB	MDBR	8.6	9.3	9.9	10.1	9.8	7.9	8.4	8.6	8.6	8.5	9.2	9.1
		MCDBR	12.2	12.1	12.1	12.6	11.4	9.0	9.7	10.0	9.8	9.3	10.4	11.0
		MCWBR	10.9	10.9	9.3	8.2	6.8	4.9	4.3	4.4	4.6	5.6	7.4	9.0
	5% WB	MCDBR	10.9	11.0	10.1	9.2	9.0	8.2	8.8	8.9	8.8	7.7	8.6	9.2
		MCWBR	11.0	11.3	10.0	8.2	7.4	5.0	4.5	4.4	4.6	5.5	7.7	9.3
Clear-Sky Solar Irradiance	taub	0.435	0.469	0.523	0.537	0.534	0.529	0.506	0.497	0.489	0.476	0.440	0.427	
	taud	2.116	2.052	1.918	1.940	2.020	2.126	2.221	2.246	2.226	2.174	2.197	2.139	
	Ebn at Noon	244	247	243	246	246	246	252	253	250	243	241	239	
	Edh at Noon	42	49	59	60	56	50	45	44	43	43	38	39	
All-Sky Solar Radiation	RadAvg	728	921	1232	1465	1527	1312	1824	1800	1455	1146	836	702	
	RadStd	70	104	89	140	167	171	180	112	153	124	94	66	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	-0.88	N/A	N/A	N/A	N/A	+14	+112	N/A	N/A
Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page